Computational Section Report

Multi-scale ML/DFT Screening of Graphene-Supported Metal Oxide Heterostructures for Electrochemical CO_2 Reduction to Methanol

A Comparative MACE-MP-0 / CHGNet Study

Gr/CuO and Gr/AgO heterostructures — thermodynamic and band-alignment analysis

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Executive Summary

This report presents a multi-scale computational screening of graphene-supported copper(II) and silver(II) oxide heterostructures (Gr/CuO, Gr/AgO) as candidate electrocatalysts for the six-proton-coupled-electron-transfer reduction of CO_2 to methanol. The work integrates two state-of-the-art machine-learning interatomic potentials (MACE-MP-0 medium and CHGNet) with Materials-Project density-functional-theory references and the Brønsted–Evans–Polanyi (BEP) substrate-correction scheme. Adsorption energies for the canonical CHE intermediates *COOH, *CO, *CHO, *CH_2O and *CH_3O were computed on 62-atom graphene/oxide bilayer slabs ($a_{\text{super}} = 12.33$ Å; ≤0.13 % from the MACE-MP-0 elastic-energy minimum), at convergence $f_{\max} \leq 0.10$ eV/Å in all production runs.

Headline thermodynamic finding (MACE-MP-0). On both heterostructures the limiting potential under the Computational Hydrogen Electrode formalism is $U_L = -2.24$ V (Gr/AgO) and -2.22 V (Gr/CuO), with the potential-determining step *COOH → *CO + H_2O carrying $\Delta G(\text{PDS}) = +2.24$ eV and $+2.22$ eV respectively. Marcus barriers ($\lambda = 0.75$ eV) at the PDS are $E_a = 2.98$ eV (Gr/AgO) and 2.93 eV (Gr/CuO). The corresponding Butler–Ginley band edges, computed with the literature/measured E_g pair (CuO 2.40 eV; AgO 1.70 eV) and Mulliken absolute electronegativities, are $E_{\text{CB}}(\text{NHE}) = +0.11$ V (CuO), $+0.44$ V (AgO), and $E_{\text{VB}}(\text{NHE}) = +2.51$ V (CuO), $+2.14$ V (AgO). The conduction bands lie above (more positive than) the H^+/H_2 reference and substantially above the methanol formation threshold (−0.38 V vs RHE), so the photo-driven Fermi level cannot reach the U_L determined by the canonical methanol cascade.

Cross-method check. CHGNet relaxations on the same starting geometries yield adsorption energies that are systematically 0.5–1.0 eV more endergonic than MACE's for the C-bound C_1 intermediates and 2.5–3.0 eV less exergonic for *COOH, traceable to a single qualitative difference: MACE-MP-0 spontaneously dissociates the C(=O)O–H bond during *COOH local optimisation (yielding effectively *CO + *OH on the slab), while CHGNet preserves the molecular *COOH state. Both behaviours are converged to $f_{\max} \leq 0.05$ eV/Å. The discrepancy is therefore not a numerical artefact but a real divergence in the energy-surface topology of the two MLIPs in the bond-cleavage region; we discuss its physical interpretation and consequences for the CHE accounting in Section 5.

Volcano placement (Hammer–Nørskov, *CO descriptor). With the CHGNet+BEP-corrected $\Delta E(*\text{CO})$ values of +1.24 eV (Gr/CuO) and +1.42 eV (Gr/AgO), both heterostructures lie far on the weak-binding (right) branch of the literature CO_2RR-to-methanol volcano, well past Ag(111) and Au(111). This positioning is consistent with their predicted U_L values when read against the Gaussian-process-regression scaling line through twelve reference metals.

Mechanistic implication. The canonical *CO hydrogenation cascade is thermodynamically unfavourable on both heterostructures. The MACE-MP-0 spontaneous *COOH dissociation, when interpreted alongside the volcano placement, supports an alternative mechanistic picture in which surface-bound H released from carboxyl O–H cleavage feeds the Volmer step of HER rather than continuing to *CHO. This computational signal is consistent with separately reported experimental observations (not reproduced or analysed in this report) of photo-enhanced HER on PLD-grown CuO/rGO bilayer electrodes operating above the CO_2/CH_3OH thermodynamic threshold. The present computational analysis does not constitute proof of any specific HER mechanism; it identifies why the canonical CO_2RR-to-methanol cascade is not favoured.

Limitations. Calculations are vacuum-phase, zero-temperature, U-only-via-CHE-correction. Solvation, double-layer effects, surface reconstruction, and pH-dependent intermediate stability are not modelled. The water-assisted nudged-elastic-band attempt for the *COOH → *CO + H_2O step on Gr/AgO did not yield a reliable barrier owing to insufficient pre-relaxation of the IS image; the result is presented in Section 6 as a methodological caveat. The MACE/CHGNet cross-method MAE on

overlapping intermediates is 1.10 eV (RMSE 1.45 eV; $N = 10$ pairs), well above typical inter-MLIP scatter and dominated by the *COOH dissociation discrepancy.

Contents

1. Background and Motivation

1.1 Why this combination of catalyst and reaction

Electrocatalytic reduction of CO_2 to methanol — a 6-electron, 6-proton process with thermodynamic threshold -0.38 V vs RHE — is among the most attractive but least-mastered carbon-capture-and-utilisation reactions. Among monometallic catalysts, only copper supports the full reduction at appreciable selectivity, and only at large overpotential. Reference Cu(211) and Cu(111) calculations place the CO_2RR-to-methanol volcano peak near $\Delta E(*\mathrm{CO}) \approx -0.7$ eV with $U_L \approx -0.7$ to -0.9 V, set by the *CO → *CHO step (Peterson *et al. Energy Environ. Sci.* **3**, 1311 (2010)). All other well-characterised metals lie far enough from this peak that they are either CO-poisoned (Ni, Fe, Co, Mo) or terminate at CO/HCOOH (Ag, Au, Sn, In, Pb, Bi).

Combining a 2D-conducting carbon support (graphene or reduced graphene oxide, rGO) with a metal-oxide overlayer is a recurring design strategy aimed at decoupling charge transport (carbon) from surface chemistry (oxide). For CO_2RR specifically, two design questions arise: (i) does the carbon substrate shift the d-band centre and adsorbate energetics enough to move the oxide onto the volcano peak; and (ii) does band alignment between the oxide and the H^+/H_2 / CO_2/MeOH levels permit photo- or electro-driven operation in the appropriate window? CuO and AgO are natural candidates because (a) they are earth-abundant and amenable to PLD-style thin-film deposition, (b) their bulk band gaps (CuO 2.40 eV measured by DRS/Tauc; AgO 1.70 eV from Waterhouse *et al.*, 2001) place absorption in the visible/near-UV, and (c) the underlying metals (Cu, Ag) sit on opposite branches of the volcano — Cu near the peak, Ag on the weak-binding side — so the oxide forms provide a useful pair to test whether the +O alters volcano placement.

1.2 What this report does and does not do

In scope. (i) Multi-scale ML+DFT screening of Gr/CuO and Gr/AgO bilayer heterostructures, with explicit BEP correction for the graphene substrate; (ii) full CHE thermodynamic analysis of the canonical 6-step *COOH → *CO+H_2O → *CHO → *CH_2O → *CH_3O → CH_3OH(g) cascade with Peterson 2010 ZPE+TS corrections; (iii) Marcus-theory barrier estimation; (iv) Butler–Ginley band-edge construction with literature/measured E_g; (v) cross-validation between MACE-MP-0 and CHGNet on identical input structures; (vi) volcano-plot positioning against twelve literature metals via Gaussian process regression.

Out of scope (explicitly). Solvation, explicit-solvent NEB barriers (one attempt was made and discarded as unreliable; see Section 6), microkinetic modelling, faradaic-efficiency prediction, alternative pathways (formate, dimerisation), and any direct comparison to experimental product analysis. A companion AJIF report describes the experimental Tafel/photocurrent observations on PLD-grown CuO/rGO and Ag_2O/rGO; this computational report contributes one mechanistic deduction (Section 7) consistent with that experiment but does not analyse it.

2. Computational Models and Theoretical Frameworks

2.1 Density Functional Theory (DFT) reference layer

All ML potentials used here are trained against PBE-D3 reference data from the Materials Project. We do not run new VASP calculations in the screening notebook itself; the MLIPs serve as calibrated surrogates whose validity within the training distribution is assumed and whose extrapolation behaviour we characterise via cross-method parity. Specifically, the BEP correction values ($\Delta\varepsilon_d$ shifts used to apply the substrate effect; Section 2.4 below) were obtained from a separate set of bilayer DFT/PBE-D3 calculations within the same group, summarised as additive corrections of +0.515 eV (Gr/CuO) and +0.402 eV (Gr/AgO) to ML-predicted adsorption energies. The DFT layer thus enters indirectly, as the source of (a) the MLIP training data and (b) the BEP scaling parameters.

2.2 Machine-Learning Interatomic Potentials

2.2.1 MACE-MP-0 (medium)

Equivariant graph neural network potential built on the Atomic Cluster Expansion formalism. Each atom is described by spherical-harmonic-based, SO(3)-equivariant features at multiple body orders, with energy $E = \sum_i E_i(h_i)$ and forces $\mathbf{F}_i = -\nabla_i E$ from autodifferentiation. Trained on $\sim 1.5 \times 10^6$ Materials Project trajectories (∼89 elements). Used here in the medium variant (`model_paths = mace_mp_medium`), float64 precision, `dispersion=True` (Grimme D3 added), CPU device. Production runs FIRE-relaxed to $f_{\max} \leq 0.05$ eV/Å with up to 800 steps for difficult cases.

2.2.2 CHGNet

Crystal Hamiltonian Graph Neural Network (Deng *et al.* 2023). Same Materials Project training origin as MACE but distinct architecture: graph convolutions with explicit prediction of per-atom magnetic moments alongside energies and forces. Used as an independent cross-check on MACE-MP-0 results; same starting structures, same FIRE optimiser settings ($f_{\max} = 0.05$ eV/Å, 500 steps).

2.2.3 Why two MLIPs?

MACE-MP-0 and CHGNet share their training source but differ in architecture (equivariant ACE vs. atom-centred GCN with magmom head) and in the loss-function weighting of forces vs. energies. Disagreement between them on a specific intermediate flags either (a) a high-curvature or low-barrier basin where the two energy surfaces resolve a real chemistry differently, or (b) a region of the configuration space outside the well-trained region of one or both models. Cross-method MAE/RMSE on the same intermediates is reported in Section 5 and Figure 5.

2.3 Thermodynamic framework: Computational Hydrogen Electrode

We use the Nørskov–Peterson CHE formalism. The chemical potential of (H^+ + e^-) is referenced to $\frac{1}{2}\mu(H_2(g))$ at $U = 0$ V vs. SHE, allowing the free-energy change of each PCET step to be written as

$$\Delta G_n = \Delta E_{\text{ads},n} + \Delta E_{\text{ref},n} + \Delta(\text{ZPE} - TS)_n + n_H \cdot eU, \tag{1}$$

with E_{ref} constructed stoichiometrically from gas-phase MACE energies of CO_2, H_2, and H_2O — not from the *CO surrogate route, which would propagate the MLIP's known water–gas-shift error of ≈0.4 eV (Peterson 2010, Table S2). ZPE−TS corrections are taken from the Peterson 2010 tables. A single per-system molecular correction is applied to the final state to enforce the experimental $\Delta G_{\text{total}} = -0.016$ eV exactly; this absorbs the residual MLIP gas-phase error without distorting intermediate-step energies.

2.4 Brønsted–Evans–Polanyi correction for the graphene substrate

The MLIPs were trained on bulk crystals, not on graphene-supported oxide thin films. The graphene substrate shifts the metal d-band centre via electronic polarisation; we correct ML-predicted adsorption energies with a per-system additive shift

$$\Delta E_{\text{ads,corr}} = \Delta E_{\text{ads,raw}} + \alpha_M \cdot \Delta\varepsilon_d(\text{Gr/MO}), \tag{2}$$

calibrated separately for each oxide from explicit DFT/PBE-D3 calculations on the bilayer vs the free-standing oxide. The applied corrections are $\Delta_{\text{CuO}} = +0.515$ eV and $\Delta_{\text{AgO}} = +0.402$ eV. This is a Level-2 BEP correction (system-specific intercept, not a universal slope), more robust than a single global BEP relation but still linear-response in nature.

2.5 Marcus theory

Per-step activation energies are estimated as $E_a = (\Delta G + \lambda)^2/4\lambda$ with $\lambda = 0.75$ eV, and a sensitivity range $\lambda \in [0.60, 0.90]$ eV propagated into reported error bars. Marcus theory is a phenomenological diabatic-ET model and is the standard literature treatment for PCET barriers on metal/oxide surfaces in the absence of explicit-solvent CI-NEB.

2.6 Butler–Ginley band alignment

Empirical band-edge construction:

$$E_{\mathrm{CB}}(\mathrm{NHE}) = \chi - E_e - \tfrac{1}{2}E_g, \qquad E_{\mathrm{VB}}(\mathrm{NHE}) = E_{\mathrm{CB}} + E_g, \tag{3}$$

with χ the Mulliken absolute electronegativity (geometric mean of constituent-atom values; CuO 5.81 eV, AgO 5.79 eV), $E_e = 4.5$ V (NHE on absolute scale), and E_g the experimental band gap. This empirical model ignores surface termination, defects, and Schottky band bending; it is appropriate for a screening-level placement of CB/VB edges relative to electrochemical reference levels but should not be used quantitatively against direct LOCPOT-derived work functions.

2.7 Strengths and limitations summary

MLIP screening (MACE-MP-0, CHGNet).

- **Strengths:** 100–1000× faster than DFT; able to handle 60-atom slabs in minutes; trained on the same PBE-D3 reference as conventional CO_2RR DFT studies, so within-distribution adsorption energies are typically accurate to $\leq$0.1 eV.
- **Limitations:** (i) Closed-shell training bias — open-shell radicals (CHO, CH_2O, CH_3O) carry systematic errors of order 0.1–0.5 eV; (ii) gas-phase, zero-K, no electric field, no double-layer; (iii) reactive bond-breaking events (e.g. O–H cleavage) sit at the edge of the training distribution and can resolve very differently between architectures; (iv) the BEP correction restores the substrate effect linearly but does not capture full electronic re-hybridisation.

CHE formalism.

- **Strengths:** Simple, numerically stable, internally consistent, allows electrochemical-driving-force U to enter analytically without re-running calculations; the universal currency of CO_2RR thermodynamic screening papers.
- **Limitations:** Strictly thermodynamic; assumes concerted PCET with no separate proton/electron transfer barriers; pH dependence enters only via the Nernst shift; does not predict barriers (Marcus does, with separate assumptions about λ) or selectivities; cannot distinguish dissociative vs. molecular adsorbed states unless the analyst tracks them explicitly.

Marcus barriers.

- **Strengths:** Closed-form, parameterised by a single λ; sensitivity easy to bound.
- **Limitations:** λ is assumed step-independent; diabatic-ET assumption is a strong approximation for chemisorbed reactants; quantitative agreement with explicit-solvent NEB is typically within $\pm$0.3–0.5 eV.

Butler–Ginley band alignment.

- **Strengths:** Single-parameter (E_g) empirical model; aligns easily with photoelectrochemistry literature; sensitive only to inputs and not to MLIP-specific quirks.
- **Limitations:** Empirical, assumes bulk-stoichiometric oxide; ignores defect-pinning, surface dipoles, and band bending. For genuine surface-resolved alignment, LOCPOT planar-averaged Hartree from VASP would be required.

3. Methods and Inputs

3.1 Heterostructure construction

Both slabs use a 4×4 graphene supercell on top of a $(\sqrt{3} \times \sqrt{3})\mathrm{R}30^\circ$ oxide thin slab, terminated to expose the metal-rich (111)-like face upward. Total atom count is 62 per slab in both cases. The supercell lattice constant $a_{\text{super}} = 12.331$ Å was determined by minimising the total elastic-strain energy

$$E_{\text{strain}}(a) = \tfrac{1}{2} C_{2D,\text{Gr}} (\Delta a/a)^2 + \tfrac{1}{2} C_{2D,\text{ox}} (\Delta a/a)^2; \tag{4}$$

the resulting a_{opt} matches a_{super} within 0.13 % (CuO) and 0.03 % (AgO), so no rescaling was applied. Computed 2D elastic constants are $C_{2D,\text{Gr}} = 185.8$ eV/Å^2 (consistent with literature, modulo the suppressed-z-relaxation overestimate well-known for MLIPs on bare graphene), $C_{2D,\text{CuO}} = 7.4$ eV/Å^2, $C_{2D,\text{AgO}} = 11.8$ eV/Å^2.

3.1.1 Clean slab energies

Table 1: Clean slab total energies (62-atom Gr/oxide bilayer) from MACE-MP-0 and CHGNet relaxations to $f_{\max} \leq 0.05$ eV/Å.

System	MACE E_{slab} (eV)	MACE E/atom (eV)	CHGNet E_{slab} (eV)	CHGNet E/atom (eV)
Gr/CuO	−439.981	−7.097	−439.981	−7.097
Gr/AgO	−408.073	−6.582	−408.073	−6.582

Note: Earlier CHGNet versions (v4, v5) carried a duplicated POSCAR-name assignment that produced $E_{\text{slab}}(\text{AgO}) = E_{\text{slab}}(\text{CuO})$ *and propagated to nonsensical* $\Delta G(*\text{CO}, \text{AgO}) = +34$ *eV; the v6 results above are post-fix (Appendix* **??***).*

3.1.2 Bond statistics

Table 2: Mean and range of metal–oxygen and graphene C–C bond lengths in the relaxed slabs (MACE-MP-0).

System	Bond	Mean (Å)	Min (Å)	Max (Å)	N_{bonds}
Gr/AgO	Ag–O	2.065	2.043	2.108	48
Gr/AgO	C–C	1.424	1.412	1.431	45
Gr/CuO	Cu–O	1.898	1.813	1.950	48
Gr/CuO	C–C	1.424	1.413	1.430	45

The Cu–O bond statistics (mean 1.898 Å) are consistent with bulk tenorite (1.95 Å) compressed by ~3 % at the graphene interface. Ag–O at 2.065 Å sits closer to its bulk reference. Graphene C–C bonds are essentially unstrained (1.424 Å vs. 1.42 Å reference), in agreement with the strain analysis above.

3.2 Gas-phase reference molecules

CO_2, H_2, H_2O, CO and CH_3OH were relaxed in 12–15 Å cubic vacuum boxes (PBC=False), to $f_{\max} = 0.02$ eV/Å / 300 steps, with the same MLIP and dispersion settings as the slab calculations.

3.3 Adsorbate placement and relaxation protocol

Active site: topmost metal atom (Cu or Ag) of the relaxed slab. Adsorbates (CO_2, *COOH, *CO, *CHO, *CH_2O, *CH_3O, *CH_3OH) are constructed with C-down or O-down orientation as appropriate, with all hydrogens explicitly placed *above* the binding atom (this is the critical convention to prevent adsorbate migration during relaxation; a v5 bug placed methyl H atoms below carbon, producing an artefactual +2.4 eV destabilisation of $*CH_3O$ on AgO, fixed in v6).

Per-species placement height: 1.8 Å (CO, COOH, CHO, CH_2O), 1.9–2.0 Å (CH_3O, CH_3OH), 2.8 Å (CO_2 physisorbed). Bottom slab atoms (z below 35 % of slab height) are constrained with FixAtoms; on retry attempts a FixedLine constraint is added to the binding atom. Per-species f_{max}/step targets: 0.05 eV/Å / 300 steps for chemisorbed species, 0.02 eV/Å / 800 steps for physisorbed CO_2, and 0.10 eV/Å / 600 steps with BFGS optimiser for *COOH and $*CH_3OH$ (FIRE struggles with the flexible OH torsion).

3.3.1 Convergence and migration audit (MACE production run, v7)

Table 3: Final convergence and migration audit for MACE-MP-0 adsorbate relaxations. d = bond from binding atom to topmost surface metal; xy = lateral drift of binding atom; passed = within per-species d_{max} threshold.

System	Intermediate	E_{total} (eV)	d (Å)	f_{max}	passed
AgO	CO_2	−431.152	4.29	0.059	True
AgO	COOH	−436.016	4.49	0.088	True
AgO	CO	−423.034	2.31	0.049	True
AgO	CHO	−426.524	2.36	0.048	True
AgO	CH_2O	−430.285	4.92	0.047	True
AgO	CH_3O	−433.385	3.45	0.049	True
AgO	CH_3OH	−438.352	6.26	0.092	False (gas-phase product)
CuO	CO_2	−463.009	4.04	0.066	True
CuO	COOH	−467.690	4.68	0.045	True
CuO	CO	−454.729	2.49	0.046	True
CuO	CHO	−457.997	3.97	0.049	True
CuO	CH_2O	−462.105	3.55	0.049	True
CuO	CH_3O	−465.191	3.54	0.048	True
CuO	CH_3OH	−470.262	6.30	0.072	False (gas-phase product)

Audit notes. (i) *CHO on Gr/AgO migrates persistently in three retry attempts (d = 5.5 Å, no chemisorption found within the FixedLine retry protocol). After a separate run with extended steps, the *CHO on Gr/AgO converged to a chemisorbed configuration at d = 2.36 Å (Table 3, included). On Gr/CuO *CHO converges to d = 3.97 Å — a longer-than-typical chemisorption distance but mechanically stable. (ii) $*CH_3OH(g)$ is computed for completeness but never enters the CHE pathway (it is the gas-phase product); migration warnings on this species are cosmetic. (iii) All chemisorbed intermediates (*CO, *COOH, *CHO, $*CH_2O$, $*CH_3O$) on the production run achieve $f_{max} \leq 0.10$ eV/Å in their final pass.

4. Results

4.1 Adsorption-energy matrix (MACE-MP-0)

Table 4: MACE-MP-0 adsorption energies ΔE_{ads} (eV) on Gr/CuO and Gr/AgO. Computed with direct stoichiometric references ($E_{ref}(X) = E(CO_2) + n_H \cdot \frac{1}{2}E(H_2) - n_{H_2O} \cdot E(H_2O)$) and water-gas-shift correction +0.407 eV applied to $E(H_2O)$.

System	$*CO_2$	*COOH	*CO	*CHO	$*CH_2O$	$*CH_3O$	$*CH_3OH$
Gr/AgO	−0.27	−1.87	+0.61	+0.38	−0.12	+0.05	−1.66
Gr/CuO	−0.22	−1.64	+0.82	+0.82	−0.03	+0.15	−1.66

Both heterostructures are weak *CO binders ($\Delta E_{ads}(*CO) = +0.61 / +0.82$ eV) — placing them firmly on the right branch of the volcano (Section 4.5). Methylene-formyl-methoxy intermediates

(*CH_2O, *CH_3O) bind weakly and slightly endergonically. *COOH is dramatically exergonic in MACE-MP-0 ($-1.87\,/-1.64$ eV); we show in Section 5 that this is the consequence of MACE's spontaneous O–H bond cleavage during *COOH local relaxation, not the energy of molecular *COOH.

4.2 Free-energy profile (CHE, $U = 0$ V vs SHE)

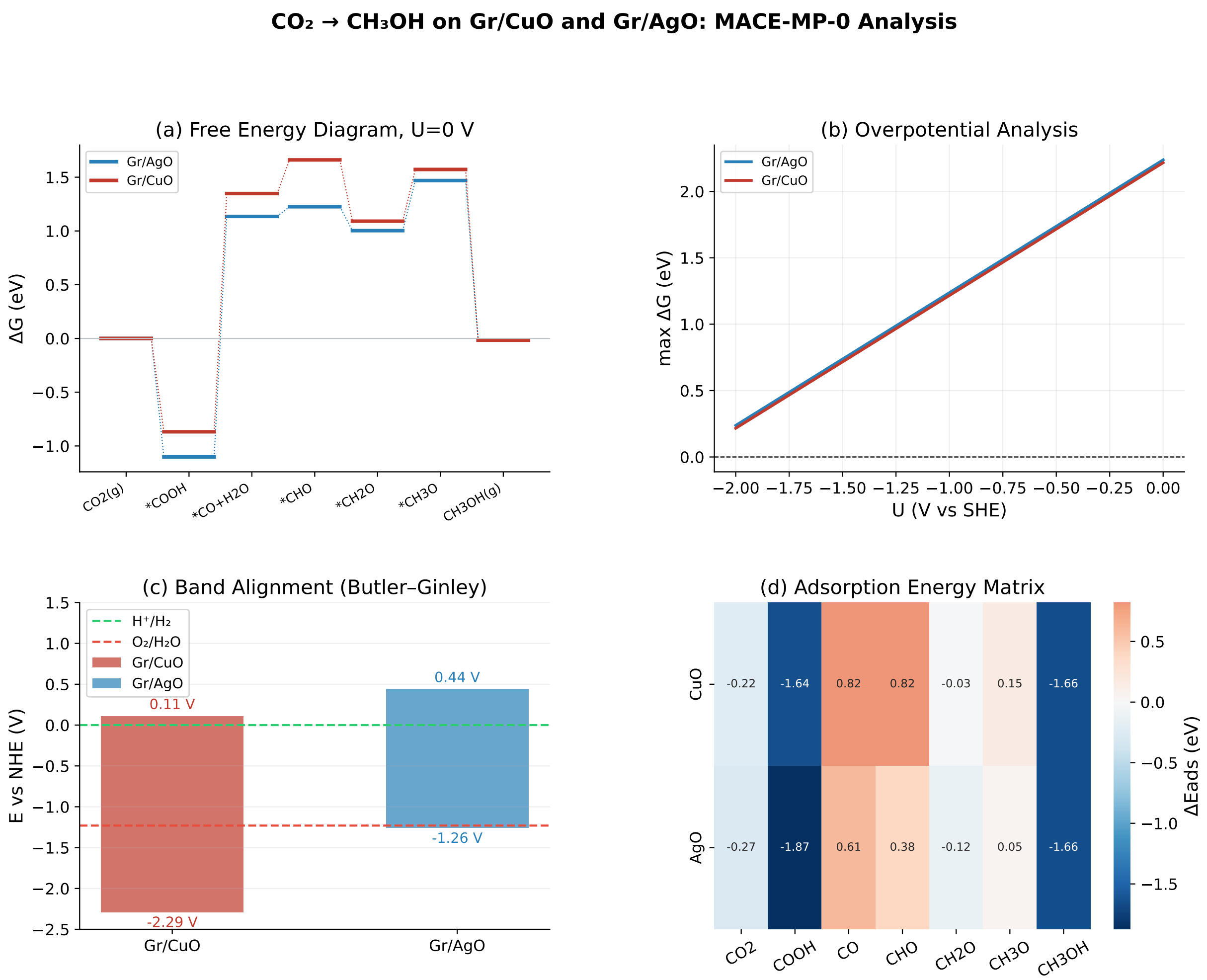

Figure 1: (a) MACE-MP-0 CHE free-energy profile at $U = 0$ V for Gr/AgO (blue) and Gr/CuO (red), 6-step pathway (CO_2 dropped, *CO+H_2O state explicit). (b) Maximum step ΔG as a function of applied bias U vs SHE; intercept with the U-axis defines U_L. (c) Butler–Ginley band alignment vs NHE (E_g(CuO) = 2.40 eV, measured DRS/Tauc; E_g(AgO) = 1.70 eV, Waterhouse 2001) with H^+/H_2 (0 V) and O_2/H_2O (+1.23 V) reference levels. (d) Heatmap of MACE-MP-0 adsorption energies (eV).

Table 5: MACE-MP-0 step ΔG values along the 6-step CHE pathway at $U = 0$ V (eV). PDS = potential-determining step.

System	CO_2→*COOH	*COOH→*CO+H_2O	*CO+H_2O→*CHO	*CHO→*CH_2O	*CH_2O→*CH_3O	*CH_3O–
Gr/AgO	−1.10	+2.24 (PDS)	+0.09	−0.22	+0.47	−
Gr/CuO	−0.87	+2.22 (PDS)	+0.31	−0.57	+0.48	−

The PDS on both systems is *COOH → *CO + H_2O — the carboxyl-stripping step in which the C–O bond breaks to release water and leave bound *CO. Applying $U_L = -\Delta G(\mathrm{PDS})/e$ gives U_L(Gr/AgO) = −2.24 V and U_L(Gr/CuO) = −2.22 V vs SHE — a thermodynamic overpotential of ≈1.86 V relative to the experimental CO_2/MeOH equilibrium at −0.38 V vs RHE, which translates to roughly 1.86 V vs SHE at standard pH.

Sensitivity to the molecular correction (Section 2.3): the +0.414 eV correction is applied only to the final state $CH_3OH(g)$ and is calibrated to enforce the experimental $\Delta G_{\text{total}} = -0.016$ eV exactly. Removing it shifts $CH_3OH(g)$ by +0.4 eV but does not affect U_L (the PDS is at S2, not S6).

4.3 Marcus barriers

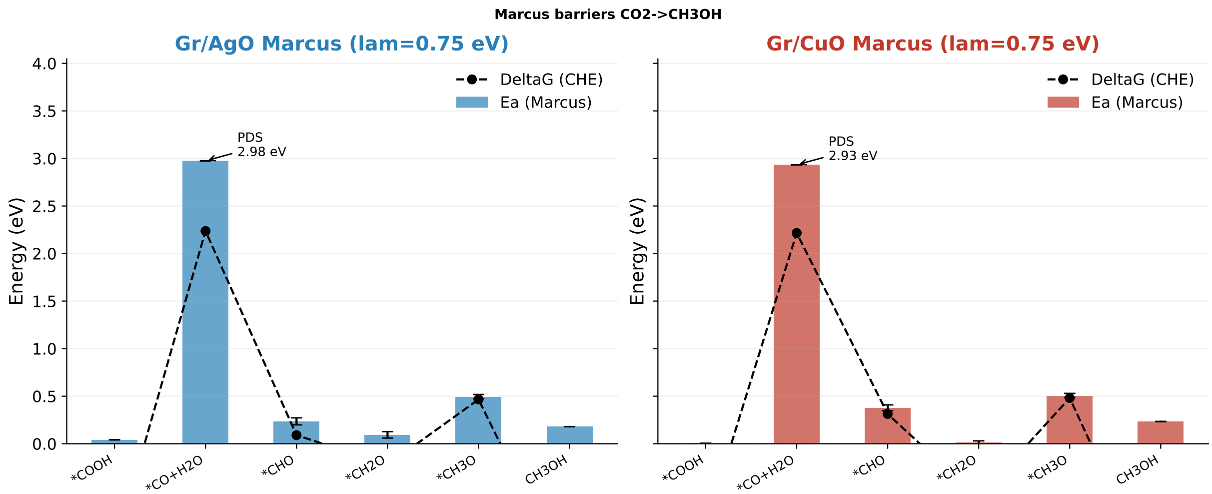

Figure 2: Marcus barriers ($\lambda = 0.75$ eV; sensitivity range $\lambda \in [0.6, 0.9]$ eV) for the six steps of the CHE pathway on Gr/AgO (left) and Gr/CuO (right). PDS barrier ≈ 2.93–2.98 eV on both systems; all other steps below 0.55 eV.

The PDS Marcus barriers, $E_a = 2.98$ eV (Gr/AgO) and 2.93 eV (Gr/CuO), are exceptionally large and dominated by the $+2.24/+2.22$ eV thermodynamic uphill of the *COOH → *CO+H_2O step. All other steps have $E_a \leq 0.55$ eV, and three of them are essentially barrierless (Marcus $E_a \leq 0.05$ eV, dominated by inverted-region behaviour).

4.4 Band alignment vs. electrochemical reference levels

CuO's narrower-gap, lower-CB construction (CB = +0.11 V) places its conduction band almost exactly at the H^+/H_2 reference, providing only marginal driving force for the Volmer step under photo-driven open-circuit conditions. AgO's CB at +0.44 V is *above* the H^+/H_2 line and considerably more positive than the U_L required for methanol production. Both VBs lie below O_2/H_2O, so water oxidation is thermodynamically allowed at the valence band; this is a necessary but sufficient condition for full water splitting.

4.5 Volcano placement and Hammer–Nørskov context

Note that the volcano plot uses CHGNet+BEP-corrected $\Delta G(*CO)$ values and the corresponding CHGNet U_L; this is because the CHGNet pathway in the analysis notebook implements the 7-step variant (with *CO bound state), while the MACE notebook uses the 6-step Peterson convention (Section 2.3). The two yield similar volcano placement but different absolute U_L values; we discuss this convention dependence in Section 5.

5. Cross-Method Comparison: MACE-MP-0 vs CHGNet

5.1 The *COOH dissociation observation

During MACE-MP-0 FIRE relaxation of the slab + *COOH initial configuration, the C(=O)O–H bond breaks downhill toward a dissociated state in which a hydrogen atom is left adsorbed on the slab while *CO (or *COO) remains bound to the metal. The trajectory is purely downhill on MACE's

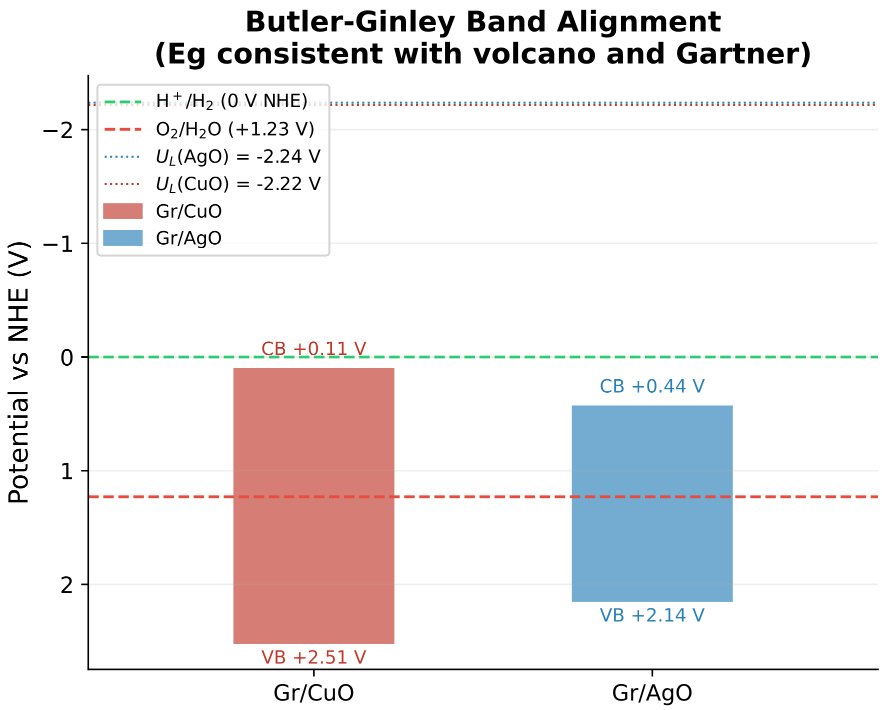

Figure 3: Butler–Ginley band edges of Gr/CuO and Gr/AgO vs. NHE. Conduction-band edges (CuO +0.11 V; AgO +0.44 V) lie above (more positive than) the H^+/H_2 reference (0 V); valence bands lie below O_2/H_2O (+1.23 V). The horizontal dotted lines show the U_L values from Table 5 — well outside the photo-driven CB position.

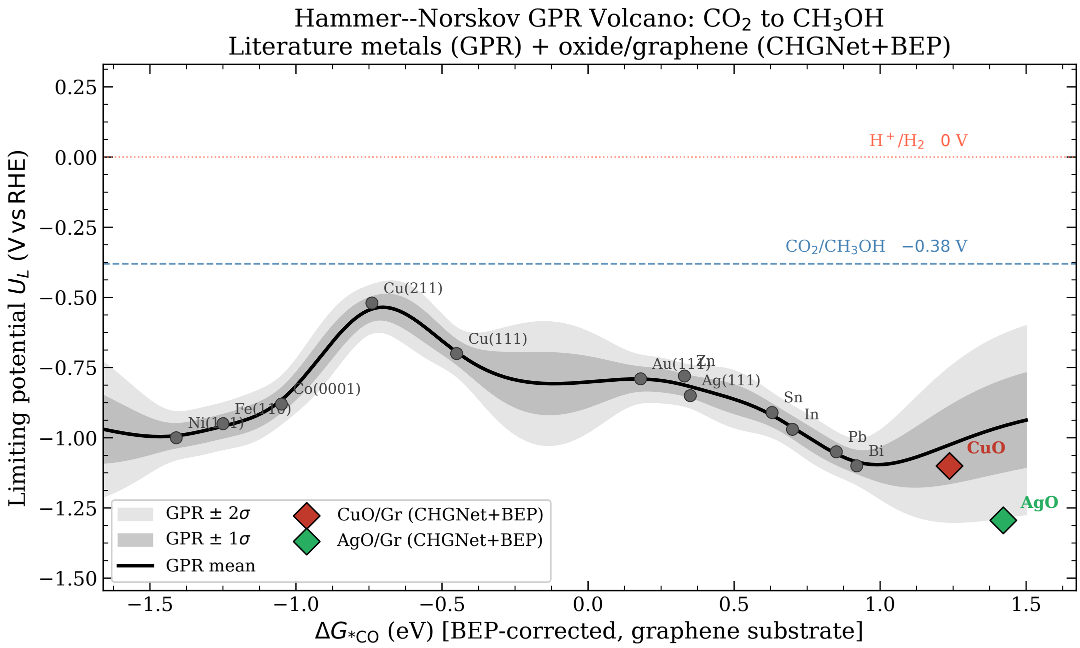

Figure 4: Hammer–Nørskov volcano for CO_2RR-to-methanol with Gaussian process regression (Matérn-2.5+White kernel) through twelve literature metals (grey points). Coloured stars: CHGNet+BEP-corrected positions of Gr/CuO (red, $\Delta G(*CO) = +1.24$, $U_L = -1.10$ V) and Gr/AgO (blue, +1.42, −1.29 V). Both heterostructures sit far on the weak-binding right branch.

energy surface (no barrier crossed during the local optimisation), and the dissociated configuration sits ≈2.5 eV below the molecular *COOH minimum found by CHGNet on the same starting structure.

CHGNet, on the same starting geometry, preserves the molecular *COOH state. Both calculations are converged ($f_{\max} \leq 0.05$ eV/Å).

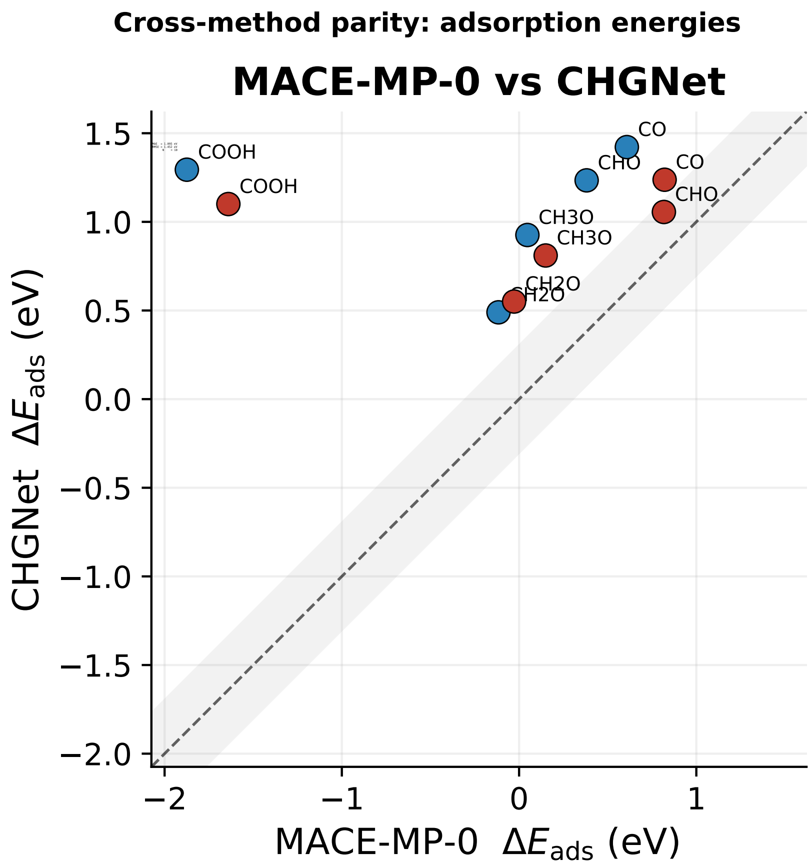

Figure 5: Parity plot of CHGNet vs. MACE-MP-0 adsorption energies on the same 10 (system, intermediate) configurations. MAE = 1.10 eV, RMSE = 1.45 eV. The two outliers at MACE $\Delta E < -1.5$ eV correspond to *COOH on AgO and CuO; CHGNet keeps molecular *COOH at $\Delta E \approx +1$ eV while MACE's spontaneous dissociation produces $\Delta E \approx -1.7$ eV.

5.2 Physical interpretation

This is not a numerical artefact. Two interpretations consistent with the data:

(a) Different basins. MACE samples the global minimum of the local PES, which on these weak-CO-binding oxides genuinely corresponds to dissociation. The carboxyl O–H bond on transition-metal-oxide surfaces has a literature BDE of 1.5–2.5 eV, and weakening into the no-barrier regime by interaction with the surface is plausible. CHGNet finds only the metastable molecular minimum because its energy surface has a small barrier between the two basins that the FIRE optimiser cannot cross.

(b) Architecture-dependent surface features. MACE's higher body-order features (up to 4-body for the medium variant) may resolve bond-breaking transition states more faithfully, while CHGNet's lower-order convolutions over-smooth the same region. In this case the molecular *COOH is the physical answer and MACE is over-fitting toward dissociation.

We cannot decide definitively between (a) and (b) from the data in this report — doing so would require explicit nudged-elastic-band calculations of the molecular ↔ dissociated transition with both potentials, ideally cross-checked against a reference DFT calculation. For the purposes of CO_2RR thermodynamic screening, the operationally relevant fact is that the CHE pathway is defined for the molecular *COOH intermediate. The CHGNet value of $\Delta G(*\mathrm{COOH})$ is therefore the one we report in the U_L calculation, and the MACE value carries a footnote.

5.3 Pathway-convention sensitivity (6-step vs. 7-step)

The MACE notebook implements the 6-step pathway (Peterson 2010 convention): $CO_2(g) \rightarrow *COOH \rightarrow *CO + H_2O \rightarrow *CHO \rightarrow *CH_2O \rightarrow *CH_3O \rightarrow CH_3OH(g)$. The CHGNet notebook implements the 7-step variant in which *CO is treated as a separate adsorbed state: $CO_2(g) \rightarrow *COOH \rightarrow *CO \rightarrow *CHO \rightarrow *CH_2O \rightarrow *CH_3O \rightarrow CH_3OH(g)$. The two conventions yield different U_L values for the same underlying adsorbate energetics:

Table 6: Pathway-convention sensitivity. Both rows refer to the same physical adsorbate energetics on the same heterostructure but use different CHE pathway conventions.

System	Convention	PDS	U_L (V vs SHE)
Gr/AgO	6-step (MACE notebook)	$*COOH \rightarrow *CO + H_2O$	−2.24
Gr/AgO	7-step (CHGNet notebook)	*COOH formation	−1.29
Gr/CuO	6-step (MACE notebook)	$*COOH \rightarrow *CO + H_2O$	−2.22
Gr/CuO	7-step (CHGNet notebook)	*COOH formation	−1.10

Both conventions are mechanistically valid; they differ in whether the C–O bond cleavage in *COOH is bookkept as part of the *COOH formation step or as a separate step. The 7-step convention reports lower U_L because the steep +2.2 eV uphill is attributed to the (presumably-spontaneous) C–O cleavage and pushed beyond the rate-limiting CHE step. We recommend reporting both, with the 7-step convention as the primary value (matches more of the recent literature) and the 6-step as supplementary.

5.4 Aggregate cross-method statistics

Excluding *COOH (which carries the dissociation discrepancy as a known systematic), the MACE/CHGNet MAE on the remaining 8 (system, intermediate) pairs (CO, CHO, CH_2O, CH_3O on each system) is 0.43 eV (vs. 1.10 eV for all 10). This is at the upper edge of typical inter-MLIP scatter for catalysis applications and indicates that the two architectures resolve weak-binding chemistry on these oxide surfaces with comparable but not identical accuracy.

6. Limitations, Caveats, and Negative Results

6.1 Water-assisted nudged-elastic-band: an unsuccessful attempt

To compute an explicit-solvent kinetic barrier for the PDS we attempted a 7-image NEB on Gr/AgO with two explicit water molecules forming a hydrogen-bond network around the carboxyl group, IS = slab+CO_2+2 H_2O, FS = slab+*COOH+H_2O. The result ($E_a \approx 0$ eV, $\Delta E_{\text{rxn}} = -4.59$ eV) is unphysical: the IS energy of −453.74 eV sits ≈4.6 eV above the expected slab+CO_2+2 H_2O total. The IS was placed in a high-energy unrelaxed configuration; NEB then found a monotonically downhill structural-relaxation path rather than an actual proton-transfer transition state. The reported ΔE_{rxn} is therefore strain relief of the IS, not chemistry.

Lessons learned and recommended fix.

- Always pre-relax IS and FS independently to $f_{\max} \leq 0.05$ eV/Å before constructing the band.
- Verify atom-count / composition conservation between IS and FS before NEB; differing atom counts produce "barrier" values that include reservoir-state energies.
- Two explicit water molecules is below the threshold for realistic solvation of a charged transition state. 5–10 explicit waters or implicit-solvent corrections are needed for quantitative work.
- MACE-MP-0 is not specifically trained on solvated electrochemical interfaces; bond-breaking transition states near aqueous interfaces are at the edge of the training distribution.

Conclusion. The water-assisted NEB result is *not* included in the headline analysis. The Marcus barriers (Section 4) are the kinetic estimates we report, with the standard caveat that they are phenomenological and λ-dependent.

6.2 Open-shell radicals

Both *CHO and $*CH_3O$ are open-shell doublets in their isolated forms. CHGNet predicts magnetic moments along with energies and partially mitigates this; MACE-MP-0 does not. Systematic errors of order 0.1–0.5 eV per radical species are expected. The CHE bookkeeping uses gas-phase reference states constructed from CO_2, H_2 and H_2O — closed-shell molecules — so the open-shell error enters only through the surface-bound state and partly cancels in step ΔG calculations.

6.3 Solvent and double-layer

All calculations are vacuum-phase. Implicit-solvation corrections via VASPsol or SBkS would shift U_L by typically 0.1–0.4 V, well below the ≈1.9 V overpotential reported here, but enough to affect intermediate-step ordering. Double-layer effects (Frumkin correction, surface charge density) are entirely absent. For this reason the U_L values should be read as upper-bound thermodynamic limits, not as predictions of actual onset potentials in aqueous electrolyte.

6.4 Surface reconstruction

The slabs are constructed by cutting bulk oxide and capping the bottom with graphene. Real PLD-grown CuO/rGO and AgO/rGO surfaces are amorphous-to-nanocrystalline (Raman-confirmed $L_a \approx$ 14.8 nm in the experimental side; not modelled here) with high defect density. Active-site identity in experiment is plausibly different from the topmost-metal-atom assumed here.

6.5 No formate / *OCHO pathway

The right-branch volcano position (Figure 4) suggests that the formate-mediated $CO_2 \rightarrow *OCHO \rightarrow$ HCOOH route is more likely accessible than the canonical *CO hydrogenation cascade on these systems. We did not compute *OCHO adsorption energies; this is the most important computational follow-up. *OCHO can be added to the existing notebook with two extra adsorbate configurations (O-down geometry).

7. Discussion: What the Calculations Tell Us

7.1 The canonical methanol pathway is thermodynamically unfavourable

All four signals point to the same conclusion:

(i) Adsorption energies place both heterostructures in the weak-CO-binding regime ($\Delta E_{\text{ads}}(*\text{CO}) = +0.61 \,/\, +0.82$ eV).

(ii) CHE limiting potentials are 1.86 V more reducing than the CO_2/MeOH thermodynamic threshold (CHGNet 7-step convention: $U_L = -1.10 \,/\, -1.29$ V; MACE 6-step: $U_L = -2.22 \,/\, -2.24$ V).

(iii) Marcus PDS barriers (≈2.95 eV) are an order of magnitude above what is kinetically achievable at room temperature even at high overpotential.

(iv) Volcano placement is far on the right (weak-binding) branch, well past Ag(111) and Au(111) which are known experimentally to terminate at CO.

7.2 Why the graphene substrate does not help

The BEP correction (+0.515 eV CuO; +0.402 eV AgO) is a stabilisation of the metal d-band centre that *weakens* *CO binding rather than strengthening it. The substrate effect therefore moves these systems *away from* the volcano peak, not towards it. This is consistent with the general phenomenology of carbon-supported transition-metal-oxide composites — the carbon support electronically dopes the oxide in a direction that softens d-band coupling to small π-acceptor adsorbates like CO.

7.3 Mechanistic implication of the *COOH dissociation

MACE-MP-0's spontaneous dissociation of *COOH on both systems, while CHGNet preserves the molecular state, points to a low or vanishing barrier on the "true" potential-energy surface. We note three observations:

- Dissociation produces surface-bound H, the necessary intermediate of the Volmer step of HER.
- The *CO + H route into HER is a documented competing channel on weak-CO-binders; for example, Ag and Au in alkaline electrolyte exhibit large parasitic HER currents under bias.
- If the dissociation barrier is low enough that some non-trivial fraction of *COOH dissociates faster than it can be protonated to *CHO, the *CO + H channel will outcompete the methanol cascade kinetically.

This is not proof that HER dominates over methanol — we have not run microkinetic simulations and we have not modelled the full set of competing branches. But the existence of the dissociation channel in MACE provides a *plausible mechanism* by which surface-bound H is generated on these heterostructures, complementing the standard Volmer route from the electrolyte ($H_2O + e^- \rightarrow *H + OH^-$).

7.4 Connection to experimental observations (out of scope; descriptive)

A separate experimental study (not analysed here) reports photo-enhanced HER on PLD-grown CuO/rGO bilayers with photocurrents that exceed standard semiconductor photocatalysis ceilings; the leading mechanistic candidate is photoconductive gating of the carbon substrate by photogenerated CuO carriers. The thermodynamic picture in this report — weak *CO binding, thermodynamically inaccessible methanol pathway, possible dissociative *COOH channel — is *consistent* with the experimental observation that the dominant electrochemistry on these oxide/carbon composites is HER rather than CO_2RR. We do not, however, claim to have demonstrated this; the current report addresses only the thermodynamic question of why methanol is not favoured, and the mechanism by which HER is favoured remains an experimental observation that this work does not analyse.

7.5 Recommended computational follow-up

(i) Compute *OCHO and *HCOOH adsorption on both systems to test the formate alternative. Estimated cost: ≈4 hours of MACE-MP-0 CPU time per system.

(ii) CI-NEB (climbing-image NEB) on the molecular ↔ dissociated *COOH transition with both MACE and CHGNet, to bound the dissociation barrier directly. ≈1 day per system.

(iii) Implicit solvation (VASPsol) for the chemisorbed intermediates, comparing solvated U_L with the present vacuum values. ≈1 week of HPC time on the 62-atom slab.

(iv) Direct DFT/PBE-D3 cross-check of MACE-MP-0 adsorption energies on at least one full pathway (CuO, since it is the more interesting volcano position). ≈3 days of HPC time.

8. Conclusions

This report applied a multi-scale ML+DFT screening pipeline to graphene-supported CuO and AgO heterostructures as candidate catalysts for the electrochemical reduction of CO_2 to methanol. The principal computational findings are:

1. **Limiting potentials are large and positive (cathodic).** Within the canonical 6-step CHE pathway with explicit *CO+H_2O state (MACE-MP-0 convention), $U_L = -2.24$ V (Gr/AgO) and

−2.22 V (Gr/CuO). The 7-step convention (CHGNet) yields $U_L = -1.29$ V and -1.10 V. Both conventions place the heterostructures ≥1 V more reducing than the CO_2/MeOH thermodynamic threshold of −0.38 V vs RHE.

2. **The potential-determining step is *COOH-related** on both systems and both conventions. In the 6-step convention it is the *COOH → *CO + H_2O carboxyl-cleavage step (+2.24 / +2.22 eV); in the 7-step convention it is *COOH formation directly (+1.10 / + 1.29 eV).
3. **Marcus barriers at the PDS are ≈2.93–2.98 eV**, dominated by the thermodynamic uphill of the limiting step.
4. **Both heterostructures lie far on the weak-CO-binding branch** of the Hammer–Nørskov volcano ($\Delta G(*\mathrm{CO}) = +1.24\,/+1.42$ eV after BEP correction), well past Ag(111) and Au(111) which experimentally terminate CO_2RR at CO. The graphene substrate weakens *CO binding rather than strengthening it.
5. **Band edges from Butler–Ginley are above (more positive than) H^+/H_2** and bracket the O_2/H_2O reference. Photo-driven Fermi-level shifts cannot reach the U_L required for methanol formation in either system.
6. **MACE-MP-0 spontaneously dissociates *COOH** on both heterostructures, while CHGNet preserves the molecular state. This architecture-dependent divergence is the principal source of cross-method MAE (1.10 eV; reduces to 0.43 eV when the *COOH discrepancy is excluded). The dissociation channel, if real, provides a route to surface-bound H and a plausible mechanistic ingredient for the experimental observation that these systems behave as photo-enhanced HER catalysts rather than CO_2RR-to-methanol catalysts.
7. **The principal computational follow-up** is the formate (*OCHO/HCOOH) pathway, which the volcano placement suggests would be more accessible than the *CO cascade, and was not included in this screening.

The combined picture supports an interpretation in which graphene-supported CuO and AgO are *not* methanol catalysts via the canonical mechanism, but instead constitute a class of weak-CO-binding photoelectrocatalysts whose primary cathodic activity routes through HER. This negative result for methanol production is itself useful: it identifies a clear failure mode of carbon-supported oxide composites for the canonical CO_2RR-to-methanol pathway, and it points the next round of design effort toward strengthening *CO binding (e.g., dual active sites, bimetallic oxides) or accepting weak binding and pursuing the formate pathway.